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Enhancing soil fertility and crop yield through biodigestate, poultry manure, and NPK integration

Authors

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Abstract

The integration of biodigestate with either inorganic fertilizer or organic manure, such as poultry manure, plays a vital role in improving crop productivity. To evaluate the effects of these amendments, a screenhouse experiment was conducted to assess the impact of biodigestate, poultry manure, and NPK fertilizer on soil chemical properties, as well as the growth, yield, and mineral composition of okra (*Abelmoschus esculentus* L. Moench) and cucumber (*Cucumis sativus* L.). The study included seven treatments: biodigestate (BD), poultry manure (PM), NPK 15-15-15 fertilizer (NPK), biodigestate + poultry manure (BD+PM), biodigestate + NPK fertilizer (BD+NPK), NPK fertilizer + poultry manure (NPK+PM), and a control with no amendments. The experimental layout followed a completely randomized design (CRD) with three replications. The okra and cucumber trials were conducted simultaneously in adjacent sections of the same screenhouse. Amendments were applied at a rate of 120 kg N ha⁻¹. Findings indicated that the application of BD, PM, or NPK—either individually or in combination—significantly enhanced soil chemical properties, plant growth, yield, and mineral composition of both crops compared to the control. NPK, PM, and BD increased the fruit weight of cucumber and okra by 100.0% and 50.0%, 240.0% and 150.0%, and 196.7% and 103.3%, respectively, relative to the control. Among the sole applications, PM demonstrated the most substantial improvements in soil fertility and crop performance compared to BD and NPK. Compared to NPK and BD, PM increased the yield of cucumber and okra by 67.2% and 66.7%, and 14.0% and 25.0%, respectively. Additionally, the combination of NPK+PM (1.49 kg of cucumber and 0.12 kg of okra) resulted in better soil properties, growth, and yield than BD+NPK (0.74 kg of cucumber and 0.07 kg of okra). However, BD+PM (1.42 kg of cucumber and 0.12 kg of okra) performed comparably to NPK+PM, while exhibiting the highest nutrient composition in both okra and cucumber fruits. The BD+PM combination offers a viable alternative to NPK+PM, achieving similar yields while improving okra and cucumber nutrient content. The novelty of this study lies in its dual-crop, side-by-side assessment of BD-based nutrient management strategies under controlled conditions—a methodological approach not widely adopted in previous research. By simultaneously evaluating BD interactions with both organic and inorganic sources across two fruit vegetables, this work provides new evidence supporting BD+PM as a viable and sustainable alternative to conventional NPK-based fertilization. Future research should investigate the effects of these amendments on fruit quality attributes such as taste, shelf life, and nutrient stability.

Keywords: Biodigestate, poultry manure, NPK 15-15-15 fertilizer, okra, cucumber, soil chemical properties

Introduction

Introduction

Okra (*Abelmoschus esculentus* L. Moench) and cucumber (*Cucumis sativus* L.) are widely cultivated vegetables in tropical and subtropical regions, including Nigeria, where they contribute significantly to household nutrition and income generation. Okra is highly valued for its mucilaginous pods, which are consumed fresh in soups and stews and possess medicinal properties, including use as a plasma expander and cholesterol-binding agent (Gemedede et al., 2015). It is rich in dietary fiber, vitamin C, folate, and antioxidants (Adekiya et al., 2018), and its seeds yield oil suitable for human consumption and industrial use (Oyolu, 1983). Cucumber, on the other hand, is prized for its hydrating and medicinal qualities—serving as a natural laxative and a remedy for gastric ailments (Ernestina, 2001). Its fruits are consumed fresh, cooked, or processed, and are notable for their vitamin, mineral, and phytonutrient content (Aderinola and Abaire, 2018).

Despite their nutritional and economic importance, the productivity of okra and cucumber in Nigeria remains low. National averages fall significantly short of global yield benchmarks. For instance, Nigeria's cucumber yield stands at just 581.59 kg ha⁻¹ compared to the global average of 15–22 t ha⁻¹ (Liu et al., 2020; Okafor and Yaduma, 2021). Similarly, okra yields, which peaked at 27,275 kg ha⁻¹ in 2010, dropped sharply to 8,735 kg ha⁻¹ by 2011 (FAOSTAT, 2020). This yield gap is largely attributed to nutrient-deficient soils, particularly Alfisols, which dominate Nigeria's agricultural landscape. These soils are inherently low in organic matter,

susceptible to nutrient leaching, and require regular nutrient replenishment to sustain productivity (Agbede et al., 2019).

In addition to inherent soil limitations, external challenges such as erratic rainfall patterns, temperature fluctuations, and poor access to soil testing facilities contribute to low input efficiency. Blanket fertilizer recommendations, made in the absence of site-specific data, often lead to suboptimal fertilizer use and poor returns on investment. Moreover, the high cost and limited availability of mineral fertilizers further constrain their use among smallholder farmers.

To address these constraints, attention is shifting toward the use of organic and recycled nutrient sources as sustainable alternatives to conventional fertilizers. Organic amendments such as poultry manure (PM) have long been recognized for their ability to improve soil structure, enhance microbial activity, and supply essential nutrients (Daassi et al., 2024). However, their slow nutrient release limits immediate crop uptake. In contrast, biodigestate (BD)—a nutrient-rich by-product of anaerobic digestion of organic residues—contains both organic and mineralized nutrients that are readily available for plant uptake (Vishwakarma et al., 2023), positioning it as an intermediary between organic and inorganic fertilizers.

Although studies have shown the potential of BD to improve soil fertility and crop productivity (Dahunsi et al., 2021; García-López et al., 2023), most have focused on its sole application or its combination with chemical fertilizers (Adekiya et al., 2024a, 2024b). There is a paucity of research exploring the synergistic effects of BD when combined with other organic inputs such as PM, despite their differing nutrient release patterns and microbial interactions. Such combinations could potentially balance short-term nutrient availability with long-term soil health improvements, offering a more sustainable nutrient management strategy.

Furthermore, most previous research has emphasized either soil health improvements or crop yield responses in isolation, without considering how integrated nutrient inputs affect both yield and fruit nutritional quality, especially for dual crops like okra and cucumber cultivated under the same environmental conditions.

A critical knowledge gap exists in understanding the interactive effects of BD, PM, and NPK fertilizer—used individually and in combinations—on soil chemical properties, crop growth, yield, and nutrient composition. In particular, it remains unclear whether combining two organic sources (BD + PM) can match or outperform traditional inorganic-organic combinations (e.g., NPK + PM) in terms of both productivity and fruit quality under controlled tropical conditions. Therefore the objectives of this study are to evaluate the individual and combined effects of biodigestate, poultry manure, and NPK fertilizer on soil chemical properties, plant growth, yield, and mineral composition of okra and cucumber in a greenhouse setting. Based on these objectives, it was hypothesized that the combined application of biodigestate and poultry manure will enhance soil chemical properties, crop growth, fruit yield, and nutrient composition of okra and cucumber more effectively than either amendment used alone or in combination with NPK fertilizer.

Materials and methods

Experimental layout and design

The experiment was conducted in the greenhouse of Bowen University, Iwo, Osun State, Nigeria (coordinates: 7.6236°N, 4.1890°E; altitude: 312 m above sea level). The soil, classified as sandy loam, is derived from fine-grained granite gneiss and schist, and is correlated with the Egbeda series (plinthic), as reported by Smyth and Montgomery (1962) in their study of soils in

southwestern Nigeria. The soil in this area is also similar to those described as Plinthic Kandiudalf by Ajiboye et al. (2022), following the USDA classification (USDA, 2010), or as Luvisol according to FAO (1998).

The region experiences a bimodal rainfall pattern, with an annual average of approximately 1300 mm, and maintains an average temperature of 32°C. The climate is classified as subhumid. This study was designed to evaluate the effects of biodigestate, poultry manure, NPK 15-15-15 fertilizer, and their combinations on soil chemical properties, as well as on the growth, yield, and mineral composition of okra and cucumber.

The screenhouse, measuring 24 m × 8 m, was constructed with a galvanized iron framework, netting to protect against insect pests, a UV-resistant roof cover, and a gravel floor.

The experiment employed a completely randomized design (CRD) with a single-factor structure consisting of seven treatments: biodigestate (BD), poultry manure (PM), NPK 15-15-15 fertilizer (NPK), BD + PM, BD + NPK, NPK + PM, and a control (no fertilizer). Each treatment was replicated three times, resulting in a total of 21 experimental units per crop (okra and cucumber).

The okra and cucumber trials were conducted simultaneously within the same screenhouse. Plots for each crop were laid out side by side in separate but adjacent sections to prevent cross-contamination and to maintain crop-specific management. Within each crop section, treatments were randomly assigned to plots to ensure that each treatment had an equal chance of appearing in any replicate position. This randomization minimized bias and controlled for variability within the screenhouse environment.

To ensure valid comparisons among the different fertilizer treatments—BD, PM, NPK, and their combinations—a uniform nitrogen application rate of 120 kg N ha⁻¹ was used across all treatments. This approach ensured nutrient equivalency, allowing observed differences in soil chemical properties, plant growth, yield, and crop quality to be attributed to the type and form of nutrient sources rather than variations in nutrient quantity. This nitrogen rate (120 kg N ha⁻¹) was selected based on previous studies indicating its effectiveness in enhancing okra and cucumber productivity in Nigerian soils (Muhammad et al., 2019; Iwalewa and Amujoyegbe, 2019).

Sample Preparation

Soil samples were randomly collected from a depth of 0 to 15 cm around the Research Farm using a spade. The collected soil was mixed, sieved through a 2 mm mesh to remove stones and debris, and then 6 kg of the sieved soil was placed into perforated poly bags sized 30 × 17 cm to ensure aeration and drainage. These poly bags had holes punched in the bottom to facilitate air and water flow and were placed randomly in the screen house for cultivating cucumber and okra plants. Each grow bag contained one healthy plant, with 5 poly bags constituting a treatment, and there were 35 plants per replicate at a spacing of 60 cm between grow bags and 90 cm between row.

Application of amendments, Sowing of cucumber and okra Seeds, and Other Agronomic Practices

The amendments were applied based on 120 kg N ha⁻¹. The N in the BD, PM and NPK fertilizer were 1.5%, 2.5%, and 15% respectively (Table 2).

$$BD = \left(\frac{120}{1.5} \right) \times 100 = 8000 \text{ kg ha}^{-1} \quad 1$$

But 1 ha of land contains 2,000,000 kg of soil as standard (Agbede, 2009), it then means 1 ha = 2,000,000 kg of soil = 8000 kg of BD .

Therefore, $6 \text{ kg soil} = \left(\frac{8000}{2,000,000} \right) \times 6 = 0.024 \text{ kg or } 24 \text{ g of BD}$ 2

Following the above, PM was applied at 14.4 g per 6 kg soil; NPK was applied at 2.4 g per 6 kg soil. The combinations were applied by applying half of the single applications. Viz: BD +PM = 12 g BD + 7.2 g PM; BD+NPK = 12 g BD + 1.2 g NPK; NPK + PM = 1.2 g NPK + 7.2 g PM and control - no modification made.

Experimental procedures

The treatments—poultry manure (PM) and biodigestate (BD)—were thoroughly mixed into the soil using a hand trowel, following the method of Adekiya et al. (2024a). After mixing, a one-week waiting period was observed before planting to allow for stabilization of the amendments, as recommended by Okalebo et al. (2002).

Seeds of okra (variety NHAe-47-4) and cucumber (variety Market More) were sown at a rate of three seeds per poly bag and later thinned to one healthy seedling per bag one week after germination, consistent with procedures outlined by Adekiya and Agbede (2009). Watering commenced immediately after planting and was carried out manually every evening throughout the experiment to maintain consistent soil moisture, as recommended in tropical vegetable production protocols (AVRDC, 2003).

NPK 15-15-15 fertilizer was applied two weeks after planting using the ring method, placing the fertilizer 5 cm away from the plant base, as described by Adekiya and Agbede (2009). Weed control was performed manually by hand-pulling as needed, and insect pests were controlled

using cypermethrin applied at the manufacturer's recommended rate, following guidelines provided in the FAO Manual on Integrated Pest Management (FAO, 2021).

Soil analysis

Initial soil samples were sieved to a 2 mm particle size before the application of any amendments. At the end of the experiment, additional soil samples were collected from each bag for chemical analysis, excluding physical (mechanical) properties.

Soil texture analysis

The Bouyoucos hydrometer method was used to determine soil texture. Soil samples were pretreated with sodium hexametaphosphate to disperse aggregates. The density of suspended particles was measured at set intervals using a hydrometer, based on their differential settling rates in a dispersing solution.

Organic carbon estimation

Organic carbon content was estimated using the Walkley-Black oxidation method. Soil organic matter was oxidized with potassium dichromate ($K_2Cr_2O_7$) and concentrated sulfuric acid. The residual dichromate was titrated with ferrous sulfate to quantify organic carbon.

Total nitrogen determination

Total nitrogen was determined using the micro-Kjeldahl method. Soil samples were digested with concentrated sulfuric acid and a catalyst mixture to convert organic nitrogen to ammonium sulfate. The resulting ammonia was distilled and titrated with a standard acid.

Available phosphorus extraction

Available phosphorus was extracted using Bray 1 extractant (0.03 N NH_4F + 0.025 N HCl). The extracted phosphorus was analyzed colorimetrically after reacting with ammonium molybdate and ascorbic acid using a spectrophotometer.

Exchangeable cations analysis

Exchangeable potassium (K), calcium (Ca), and magnesium (Mg) were extracted using neutral 1N ammonium acetate (NH_4OAc). Potassium was measured using flame photometry, while calcium and magnesium were analyzed using atomic absorption spectrophotometry (AAS).

Analysis of manures and mineral composition of okra and cucumber fruits

The nutrient composition of PM and BD was assessed after sieving through a 2 mm mesh. The analyses included total N, P, K, Ca, Mg, Cu, Fe, Mn, and Zn, following procedures outlined by the Association of Official Analytical Chemists (AOAC, 2003). Upon harvest, four okra fruits and one cucumber fruit of uniform size were collected from each grow bag, and their mineral contents were determined using AOAC (2003) protocols. Each 1 g fruit sample was digested using a mixture of HNO_3 , H_2SO_4 , and HClO_4 in a 7:2:1 (v/v/v) ratio, and the mineral concentrations were quantified via AAS.

Determination of growth and yield parameters

For each treatment, growth parameters including plant height, leaf area, number of leaves per plant, and stem diameter were measured at mid-flowering—48 days after planting for okra and 40 days for cucumber. Okra plant height and cucumber vine length were measured using a meter

rule, the number of leaves was manually counted, and stem diameter was measured with a vernier caliper. Leaf area was determined using models proposed by Omolaiye et al. (2015) for okra and Blanco and Folegatti (2005) for cucumber.

Mature okra fruits were harvested every five days. The harvested fruits were counted and weighed, and the total pod weight per poly bag was used to determine yield. Cucumber fruits were harvested twice weekly (on Sundays and Thursdays) until 65 days after sowing, with each harvest's yield recorded by counting and weighing the fruits.

Statistical analysis

The collected data on growth, yield, and fruit nutrient composition for okra and cucumber were analyzed using analysis of variance (ANOVA). SPSS software (version 21.0) was used for the ANOVA, and Duncan's Multiple Range Test (DMRT) and Least significant difference (LSD) were applied to separate means at a 5% significance level.

Results

Pre-planting soil analysis

The results of the pre-planting soil analysis are presented in Table 1. Key findings include low organic matter content, strongly acidic pH, and significantly low nitrogen levels. Additionally, available phosphorus was found to be insufficient. The exchangeable K, Ca, and Mg levels were also below optimal thresholds. In terms of texture, the soil is predominantly sandy, with minimal silt and clay components, classifying it as sandy loam.

Table 1: Initial soil characteristics of the experimental site before sowing of okra and cucumber

Property	Value
Sand (%)	68.2
Silt (%)	19.1
Clay (%)	12.7
Textural class	Sandy loam
Organic C (%)	1.70
pH (water)	6.33
N (%)	0.17
P (mg/kg)	5.07
K (cmol/kg)	0.25
Ca (cmol/kg)	4.47
Mg (cmol/kg)	2.46
Na (cmol/kg)	0.26

Analysis of soil amendments used

Table 2 presents the nutrient composition of biodigestate, poultry manure, and NPK fertilizer. Among the three, NPK fertilizer had the highest concentrations of primary macronutrients—nitrogen (15%), phosphorus (15%), and potassium (15%)—consistent with its synthetic formulation. Poultry manure, although lower in NPK values compared to NPK fertilizer, exhibited significantly higher levels of secondary nutrients and micronutrients, such as calcium (3.31%), magnesium (0.91%), copper (0.42%), manganese (0.26%), zinc (5.12%), and organic matter content (50.5%). Biodigestate contained moderate amounts of nutrients, with notable levels of magnesium (1.01%) and zinc (4.20%), but generally lower NPK and micronutrient values than poultry manure. These differences in nutrient profiles likely contributed to the superior soil fertility improvement observed with poultry manure and its combinations in this study. The high organic matter content of poultry manure also suggests a greater potential for enhancing soil structure and long-term fertility compared to the other amendments/

Table 2: Nutrient values of the biodigestate, poultry manure and NPK fertilizer used

Nutrient (%)	Biodigestate	Poultry manure	NPK fertilizer
N	1.50 ± 0.03c	2.5 ± 0.1b	15 ± 2.1a
P	1.10 ± 0.02c	1.85 ± 0.1b	15 ± 2.1a
K	1.00 ± 0.02c	3.50 ± 0.2b	15 ± 2.1a
Ca	0.46 ± 0.01b	3.31 ± 0.21a	NA
Mg	1.01 ± 0.02a	0.91 ± 0.02a	NA
Cu	0.36 ± 0.01b	0.42 ± 0.01a	NA
Fe	0.15 ± 0.02b	0.17 ± 0.02 a	NA
Mn	0.14 ± 0.02 b	0.26 ± 0.01a	NA
Zn	4.20 ± 0.21b	5.12 ± 0.20a	NA
Organic matter	30.6 ± 3.1b	50.5 ± 3.6 a	NA

Note: Values followed by similar letters under the same row are not significantly different at $p < 0.05$ according to Duncan's multiple range test. NA = Not applicable

Influence of BD, PM, NPK, and their combinations on soil chemical properties of okra and cucumber

Table 3 presents the results of the impact of BD, PM, NPK fertilizer, and their combinations on the soil chemical properties of okra and cucumber. For both crops, the application of soil amendments—whether used individually or in combination—led to increased soil pH, OM, N, P, and K compared to the control. Among the sole applications, PM recorded the highest values for soil pH, OM, and N, while NPK fertilizer had the highest levels of P and K. Among the combined treatments (BD+PM, BD+NPK, and NPK+PM), BD+PM resulted in the most significant improvement in soil chemical properties. However, for nitrogen, the BD+PM treatment was not significantly different ($p < 0.05$) from NPK+PM.

Table 3: Effects of BD, PM and NPK and their combinations on soil chemical properties of okra and cucumber plants

	pH (water)		OM (%)		N (%)		P (mg/kg)		K (cmol/kg)	
	Okra	Cucumber	Okra	Cucumber	Okra	Cucumber	Okra	cucumber	Okra	cucumber
Control	5.51b	5.59b	2.21e	2.24e	0.16d	0.15d	4.88e	4.91e	0.23d	0.22d
BD	6.70a	6.68a	2.66c	2.62b	0.19c	0.18c	11.85d	10.77d	0.29c	0.28c
PM	6.73a	6.74a	3.01a	3.12a	0.21b	0.21b	22.67c	23.78c	0.31c	0.33b
NPK	5.50b	5.54b	2.22e	2.20c	0.19c	0.19c	29.7a	28.9a	0.35b	0.34b
BD+PM	6.67a	6.66a	2.69b	2.73b	0.24a	0.25a	25.80b	26.98b	0.39a	0.39a
BD+NPK	5.89b	5.91b	2.40d	2.45d	0.20b	0.21b	24.78b	24.51c	0.35b	0.34b
NPK+PM	5.91b	5.86b	2.46d	2.48d	0.24a	0.25a	26.6b	26.8b	0.35b	0.35b
SE±	0.22	0.19	0.11	0.13	0.01	0.01	3.41	3.54	0.02	0.02
LSD (0.05)	0.63	0.61	0.18	0.20	0.01	0.01	3.10	3.10	0.03	0.03

Values followed by similar letters under the same column are not significantly different at $p = 0.05$ according to Duncan's multiple range tests. BD = biodigestate; PM = poultry manure; NPK = NPK 15-15-15 fertilizer.

Effects of BD, PM, NPK, and their combinations on growth parameters of okra and cucumber

Table 4 shows the results of the effects of BD, PM, NPK, and their combinations on the growth parameters of okra and cucumber. The application of BD, PM, NPK fertilizer, and their combinations enhanced growth parameters compared to the control. Among the sole applications, PM produced the highest growth improvements. The order of effectiveness for both okra and cucumber was PM > BD > NPK. Among the combinations, BD+PM and NPK+PM improved growth more than BD+NPK. Although NPK+PM appeared to have the highest growth values, there were no statistically significant ($p < 0.05$) differences between the two combinations.

Table 4: Effects of BD, PM and NPK and their combinations on growth indices of okra and cucumber

	Plant/vine length (cm)		Stem/vine girth (cm)		Number of leaves/plants		Leaf area/plant (cm ²)	
	Okra	Cucumber	Okra	Cucumber	Okra	Cucumber	Okra	Cucumber
Control	12.6f	20.3f	2.1d	2.3c	4.3f	8.9e	4901f	5421f
BD	20.5c	37.5c	2.4c	2.6b	6.3c	10.7c	6412c	8012c
PM	22.3b	41.6b	2.7b	2.7b	6.7b	12.2b	7101b	8812b
NPK	15.5e	30.5d	2.3c	2.7b	5.0e	9.5d	5541e	6247e
BD+PM	29.0a	44.5a	2.9a	3.1a	7.4a	15.3a	7810a	9423a
BD+NPK	17.9d	32.5e	2.3c	2.7b	5.5d	10.7c	5842d	7156d
NPK+PM	30.4a	44.9a	2.8ab	3.0a	7.6a	15.6a	7890a	9501a
SE±	2.37	3.38	0.11	0.10	0.49	1.01	434.40	599.20
LSD(0.05)	1.6	1.5	0.1	0.1	0.35	0.5	300.0	400.0

Values followed by similar letters under the same column are not significantly different at $p = 0.05$ according to Duncan's multiple range test. BD = biodigestate; PM = poultry manure; NPK = NPK 15-15-15 fertilizer

Effects of BD, PM, NPK, and their combinations on the yield of okra and cucumber

The data in Table 5 illustrate the impact of BD, PM, NPK, and their combinations on yield parameters for okra (number of pods per plant and pod weight per plant) and cucumber (number of fruits per plant and fruit weight per plant). Across both crops, the application of these fertilizers—whether individually or in combination—significantly ($p < 0.05$) enhanced yield parameters compared to the control group, which recorded the lowest values. Among the individual applications, PM led to the most substantial increase in yield components. The ranking from highest to lowest was $PM > BD > NPK$. While PM significantly increased ($p < 0.05$) the number of okra pods compared to BD alone, it also increased okra pod weight by 150% and cucumber fruit weight by 240% relative to the control. Similarly, BD alone boosted okra yield by 100% and cucumber yield by 196%. NPK increased yields for both crops by 100% for okra and 103% for cucumber. Among the combinations, NPK+PM produced superior results but was not statistically ($p < 0.05$) different from BD+PM. Compared to BD+PM, NPK+PM resulted

in a further increase in cucumber yield by approximately 4.9%, while no further increase was observed for okra yield.

Table 5: Effects of BD, PM and NPK and their combinations on yield components of okra and cucumber

	Number of pods/fruits		Pod/fruit weight (kg)	
	Okra	Cucumber	Okra	Cucumber
Control	4d	1e	0.04f	0.30f
BD	6c	3c	0.08c	0.89c
PM	7bc	4b	0.10b	1.02b
NPK	6c	2d	0.06e	0.61e
BD+PM	9ab	6a	0.12a	1.42a
BD+NPK	6c	3c	0.07d	0.74d
NPK+PM	10a	6a	0.12a	1.49a
SE±	0.71	0.52	0.02	0.16
LSD (0.05)	0.9	0.8	0.009	0.12

Values followed by similar letters under the same column are not significantly different at $p = 0.05$ according to Duncan's multiple range test. BD = biodigestate; PM = poultry manure; NPK = NPK 15-15-15 fertilizer

Effects of BD, PM, NPK, and their combinations on the mineral content of okra and cucumber

The findings on the impact of BD, PM, NPK (15-15-15) fertilizer, and their combinations on the mineral content of okra and cucumber are presented in Figures 1–5. The application of BD, PM, NPK fertilizer, and their combinations significantly ($p < 0.05$) enhanced the mineral composition of okra and cucumber compared to the control group. Among the individual applications, PM yielded the highest levels of K (Fig. 1), Ca (Fig. 2), Fe (Fig. 3), Zn (Fig. 4), and Cu (Fig. 5). However, Zn content showed no significant difference between PM and BD for both crops (Fig. 4). For combined applications, BD+PM consistently resulted in the highest levels of K, Ca, Fe, Zn, and Cu in both okra and cucumber. This combination outperformed all other treatments in terms of overall mineral enrichment.

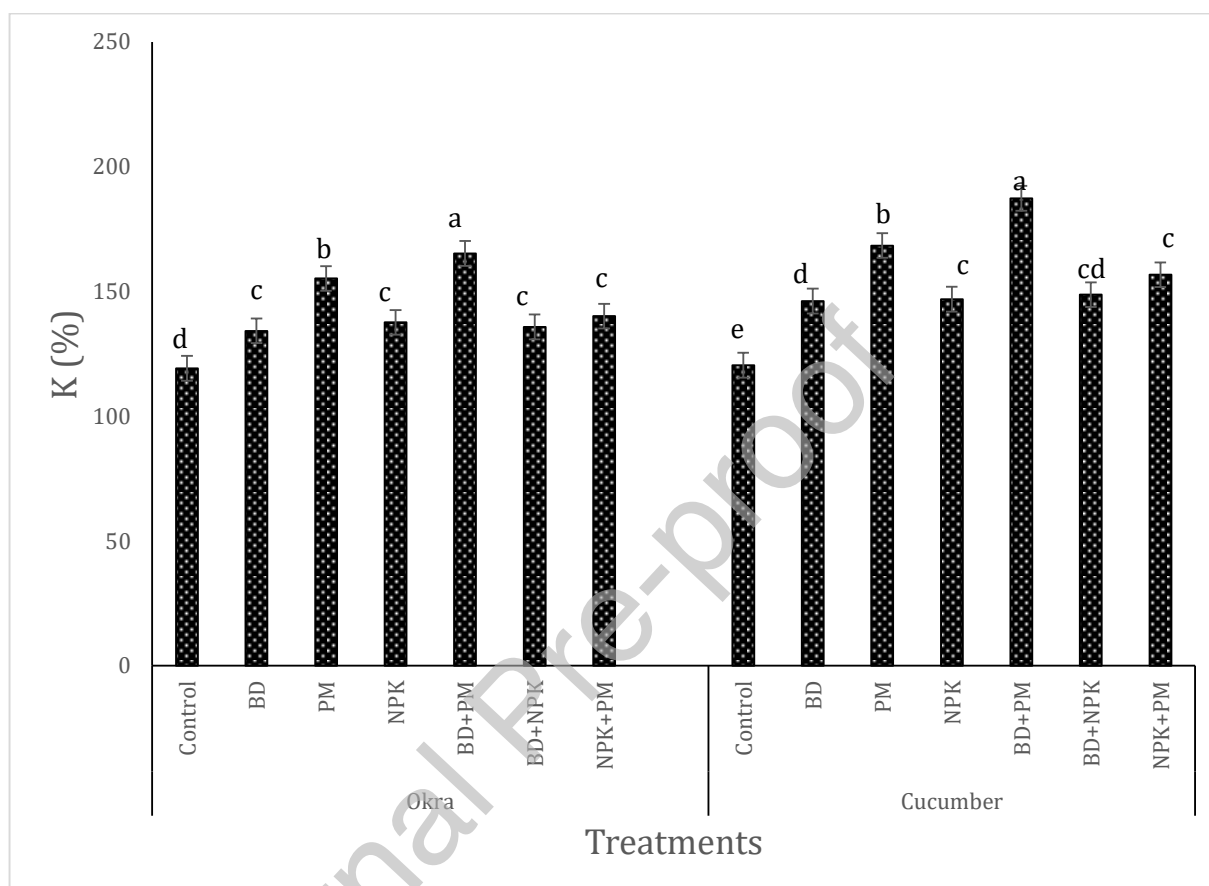


Figure 1: Effects of biodigestate (BD), poultry manure (PM), NPK 15-15-15 fertilizer (NPK), and their combinations on potassium (K) content in okra and cucumber fruits. Treatments include BD, PM, NPK, BD+PM, BD+NPK, NPK+PM, and a control (no amendment). Vertical bars represent standard errors of the means. Columns marked with different lowercase letters indicate statistically significant differences among treatments at $P < 0.05$, based on Duncan's Multiple Range Test (DMRT). Treatments sharing the same letter are not significantly different.

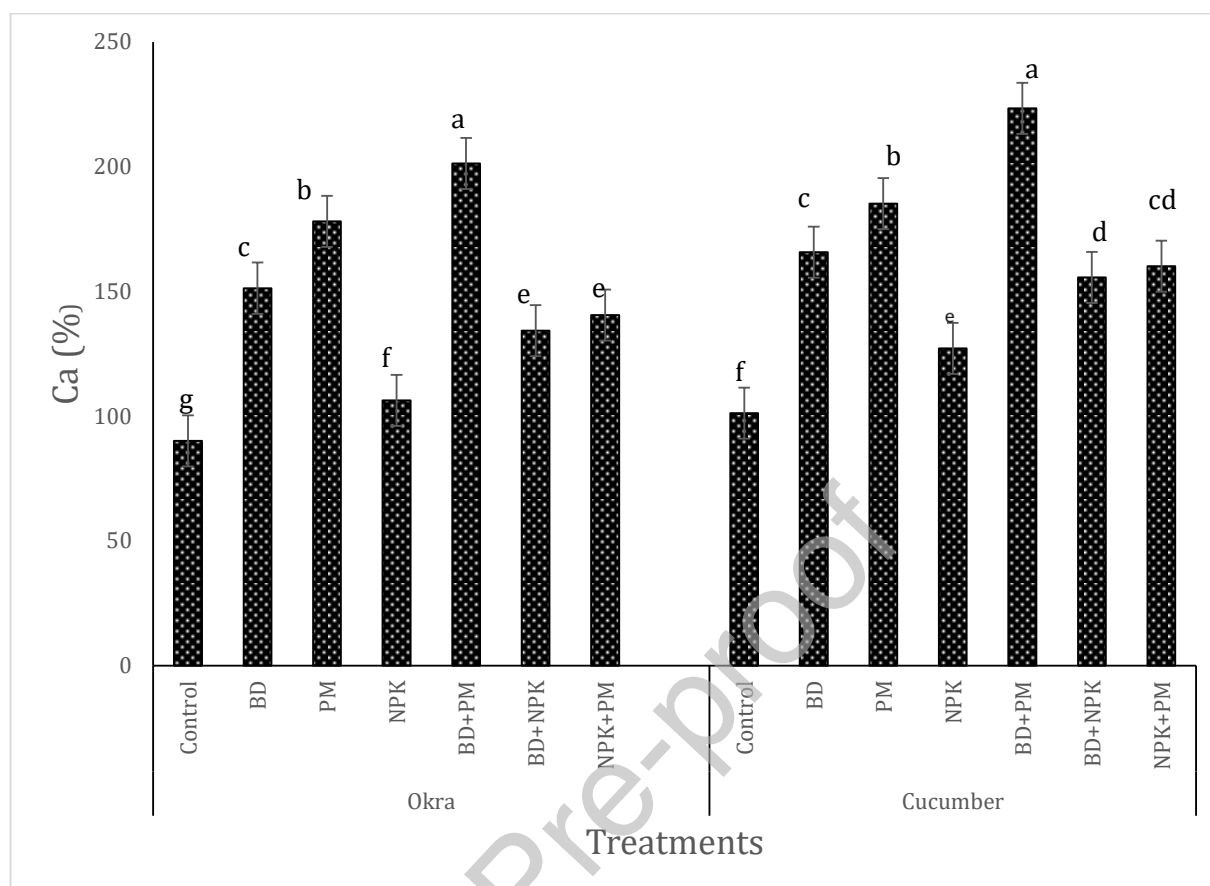


Figure 2: Effects of biodigestate (BD), poultry manure (PM), NPK 15-15-15 fertilizer (NPK), and their combinations on calcium (Ca) content in okra and cucumber fruits. Treatments include BD, PM, NPK, BD+PM, BD+NPK, NPK+PM, and a control (no amendment). Vertical bars represent standard errors of the means. Columns marked with different lowercase letters indicate statistically significant differences among treatments at $P < 0.05$, based on Duncan's Multiple Range Test (DMRT). Treatments sharing the same letter are not significantly different.

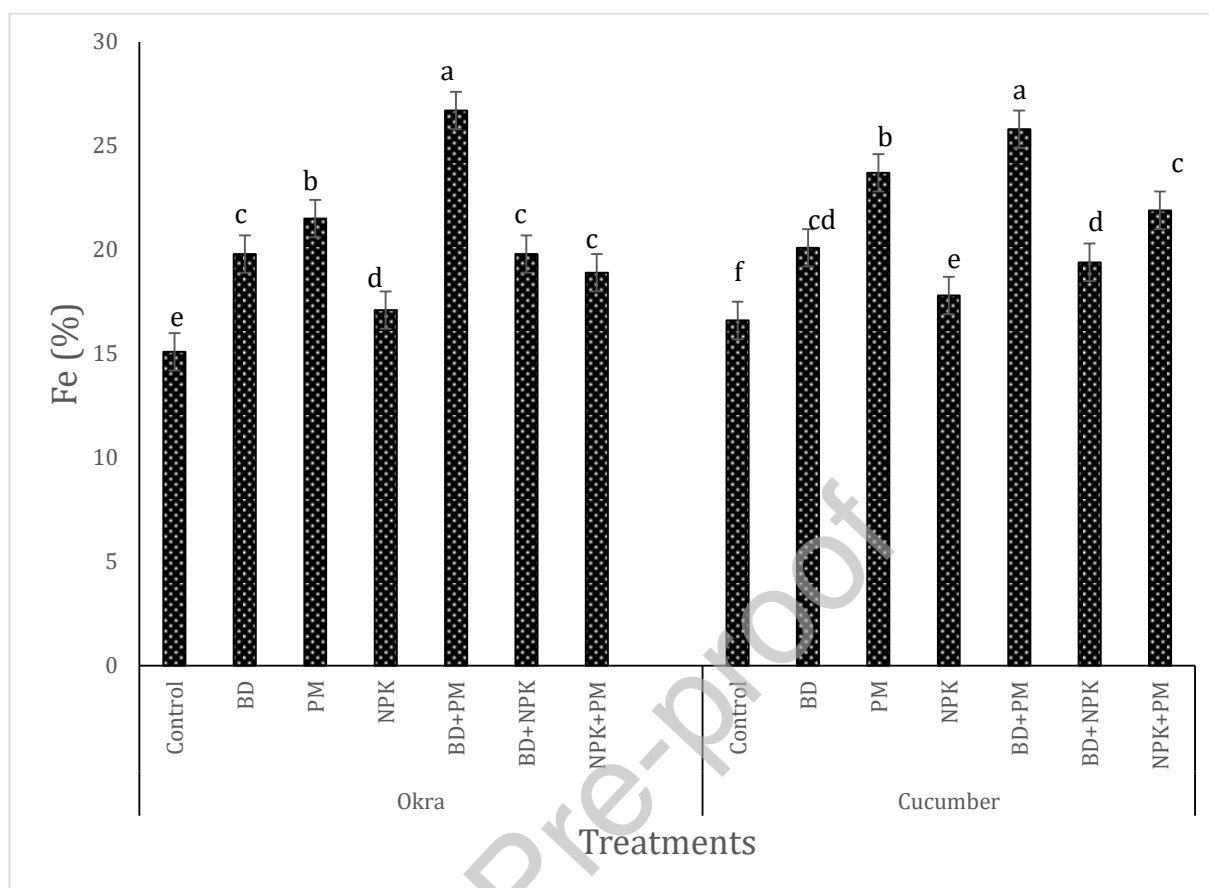


Figure 3: Effects of biodigestate (BD), poultry manure (PM), NPK 15-15-15 fertilizer (NPK), and their combinations on iron (Fe) content in okra and cucumber fruits. Treatments include BD, PM, NPK, BD+PM, BD+NPK, NPK+PM, and a control (no amendment). Vertical bars represent standard errors of the means. Columns marked with different lowercase letters indicate statistically significant differences among treatments at $P < 0.05$, based on Duncan's Multiple Range Test (DMRT). Treatments sharing the same letter are not significantly different.

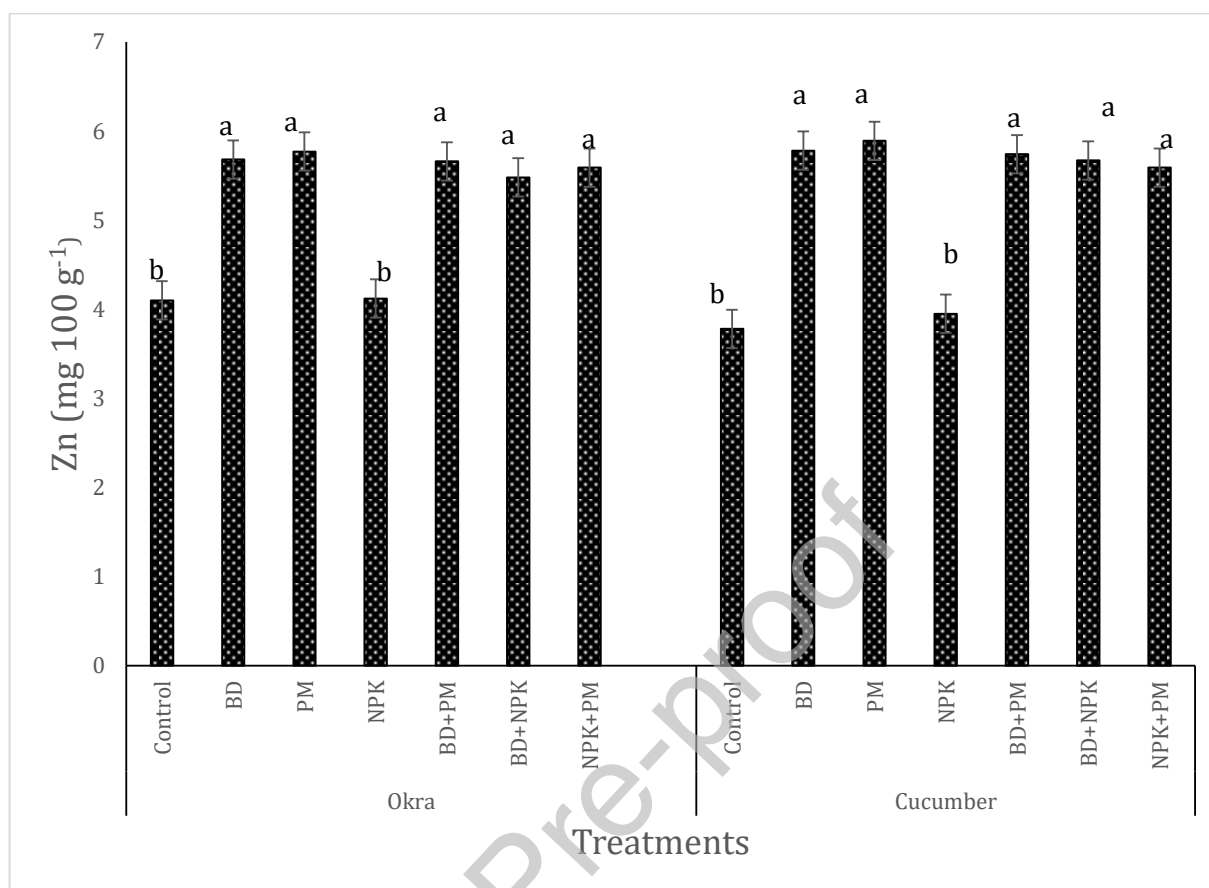


Figure 4: Effects of biodigestate (BD), poultry manure (PM), NPK 15-15-15 fertilizer (NPK), and their combinations on zinc (Zn) content in okra and cucumber fruits. Treatments include BD, PM, NPK, BD+PM, BD+NPK, NPK+PM, and a control (no amendment). Vertical bars represent standard errors of the means. Columns marked with different lowercase letters indicate statistically significant differences among treatments at $P < 0.05$, based on Duncan's Multiple Range Test (DMRT). Treatments sharing the same letter are not significantly different.

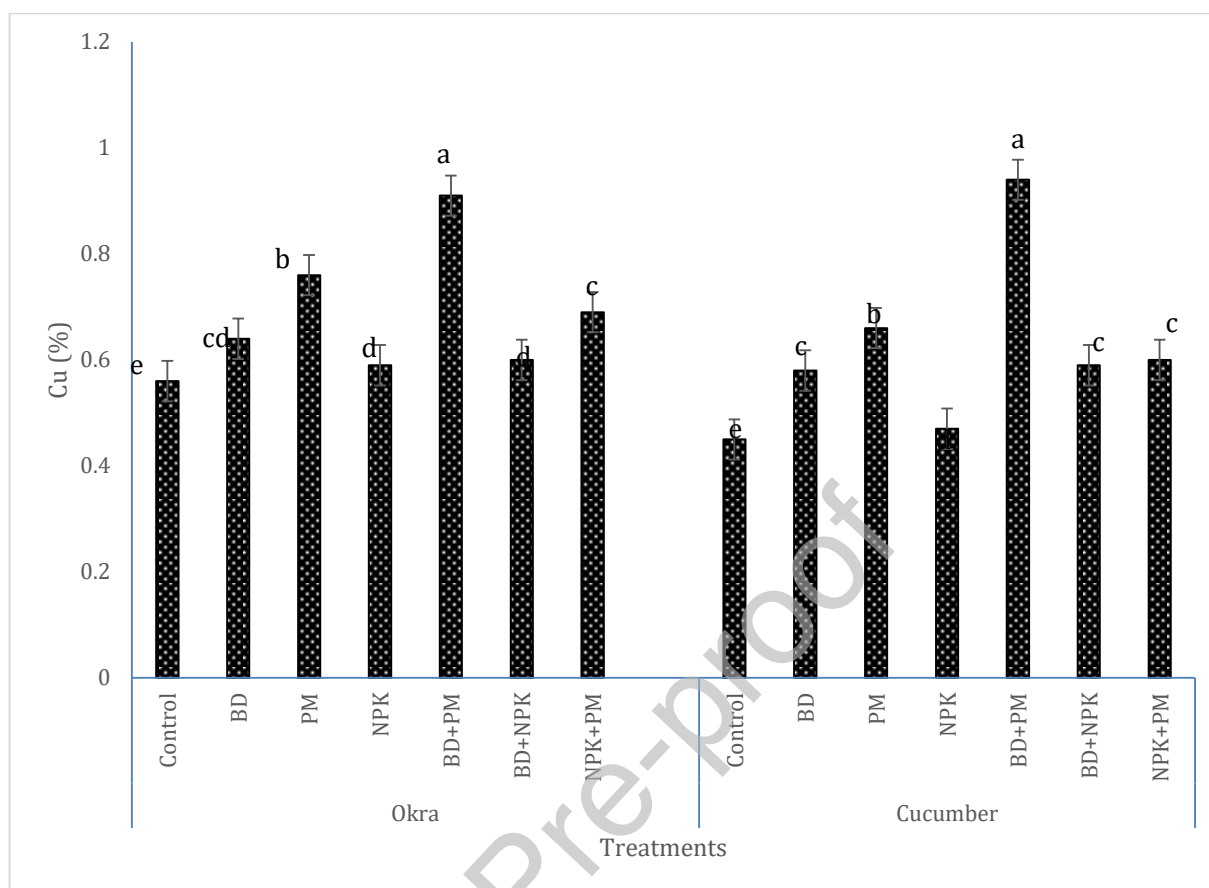


Figure 5: Effects of biodigestate (BD), poultry manure (PM), NPK 15-15-15 fertilizer (NPK), and their combinations on copper (Cu) content in okra and cucumber fruits. Treatments include BD, PM, NPK, BD+PM, BD+NPK, NPK+PM, and a control (no amendment). Vertical bars represent standard errors of the means. Columns marked with different lowercase letters indicate statistically significant differences among treatments at $P < 0.05$, based on Duncan's Multiple Range Test (DMRT). Treatments sharing the same letter are not significantly different.

Discussion

The use of BD, PM, and NPK, either alone or in combination, significantly enhanced soil characteristics. This enhancement included increases in pH, organic carbon (OC), N, P, and K levels relative to the control. This improvement is attributed to the initially low nutrient content of the soil prior to the addition of amendments (Table 1). Both BD and PM are rich in vital nutrients (Table 2), which are gradually released into the soil and absorbed by plants. The application of PM and BD improves soil chemical properties by supplying essential nutrients,

increasing organic matter content, stimulating microbial activity, raising soil pH, enhancing cation exchange capacity (CEC), ensuring steady nutrient release, and promoting microbial diversity. These changes result in more fertile and healthier soils compared to the control, which lacked organic inputs. Studies by Zaidi et al. (2023) and Adekiya et al. (2024a) report that organic amendments such as PM and BD can moderate soil acidity, often increasing pH in acidic soils due to their buffering capacity. Biodigestate, in particular, contains alkaline compounds (e.g., bicarbonates and carbonates) that neutralize soil acidity (Midden et al., 2023). Poultry manure has similar effects due to its high calcium (Ca) and magnesium (Mg) content (Masud et al., 2020).

The application of BD and PM increases soil organic matter, contributing directly to soil organic carbon pools (Daassi et al., 2024; Adekiya et al., 2024a). While NPK has minimal direct impact on OC, its combination with organic amendments stimulates microbial activity and root growth, thereby indirectly improving OC. BD, being partially decomposed, often provides readily available carbon sources that enhance microbial biomass and SOC stabilization (Barlóg et al., 2020). BD and PM are well-documented for releasing slow but sustained amounts of N, P, and K, complementing the immediate nutrient availability from NPK (Haque et al., 2023; Robles-Aguilar et al., 2019). Combined applications often exhibit synergistic effects—increasing nutrient use efficiency and reducing nutrient losses through leaching or volatilization (Wang et al., 2025). Adekiya (2019) showed that nutrient release from BD and PM is more synchronized with crop demand than NPK alone.

The BD + PM treatment outperformed or matched other treatments in improving soil chemical properties due to mechanistic complementarity: BD + PM significantly increased organic matter

content (2.69–2.73%), second only to PM alone. This relatively high OM level creates a favorable environment for microbial proliferation, which in turn enhances mineralization of organic nutrients and improves nutrient cycling. The microbial activity stimulated by PM also aids in the decomposition of labile carbon in BD, facilitating nutrient availability. The BD + PM treatment likely achieves a more balanced carbon-to-nitrogen (C:N) ratio, preventing nitrogen immobilization that may occur with high C materials alone. The combined treatment also reduces the risk of N volatilization (often seen with BD) and leaching (common with NPK), thereby enhancing nitrogen retention in the soil, making the BD + PM combination a promising sustainable alternative to inorganic fertilizer regimes for improving soil health in tropical agroecosystems. Similar increases in soil macronutrient content have been documented in other studies. Thomas et al. (2019), Zhao et al. (2021), and Dhaliwal et al. (2023) all reported that the application of organic materials resulted in higher nutrient levels in the soil. Wang et al., (2014) also reported that application of a mixture of green waste compost, sedge peat and furfural residue (1:1:1 by volume) and furfural residue substantially increased the organic carbon content of the soil and the availability of N, P, and K. This result is also in agreement with that of Agbede and Oyewumi (2023) who reported that biochar and poultry manure combinations increased soil organic matter, N, P, K, Ca and Mg contents of the soil relative to the control and their sole applications.

Additionally, the application of NPK enhanced soil fertility compared to the control, which is attributed to the high concentrations of nitrogen, phosphorus, and potassium in NPK 15-15-15 fertilizer.

The application of soil amendments substantially boosted the growth and yield of okra and cucumber plants compared to the control. PM significantly improved the growth and yield of these crops, primarily by increasing nutrient availability, improving microbial activity, and enhancing soil structure. Several studies confirm that PM is a superior organic amendment due to its high nutrient concentrations—particularly of N, P, and K—its faster decomposition rate relative to other manures, and its ability to improve soil physical properties such as water-holding capacity and structure (Howe et al., 2024). These results are consistent with the findings of Enyioko et al. (2023), who reported that poultry manure (PM) increased the yield of bell pepper under both greenhouse and open field conditions in Oforola, Owerri, Nigeria. Similarly, Ayeni and Adetunji (2010) and Ibeh et al. (2019) observed that PM significantly enhanced the growth and yield of maize and okra compared to the control. Ayoola and Adeniyani (2006) also reported that poultry manure significantly increased soil nitrogen and phosphorus levels, thereby improving yield components in cassava/maize/melon intercropping systems. Furthermore, Oke et al. (2020) found that integrating poultry manure with reduced rates of inorganic fertilizer enhanced cucumber growth and yield.

The beneficial effects of NPK fertilizer on the growth and yield of okra and cucumber were due to the initially low nutrient levels in the experimental soil, as noted by Ogbonna and Nweze (2012) and that NPK fertilizer provides immediate nutrient availability to support rapid plant growth and yield formation. Nitrogen in NPK supports leaf development and is crucial for protein and chlorophyll synthesis. Phosphorus aids in root growth, energy transfer, and cellular division, while potassium promotes stem growth, cell division, carbohydrate development and transport, and root formation. The observed enhancement in okra and cucumber growth and yield with NPK fertilizer application aligns with previous research highlighting the efficacy of

inorganic fertilizers in vegetable production. Amina et al. (2023), in a study examining various rates (0, 25, 50, 75, and 100 kg ha⁻¹) of NPK 15-15-15, reported significant increases in okra growth and yield compared to the control, with the 100 kg ha⁻¹ rate producing the best results. Similarly, Omotoso and Shittu (2007) found that NPK fertilizer significantly improved okra growth and yield parameters. Supporting this trend, Adebayo et al. (2021) assessed the effects of four NPK formulations—control, NPK 15-15-15, NPK 20-10-10, and NPKMg 12-12-17-2—on cucumber. Their results showed that NPK fertilizers significantly ($P \leq 0.05$) improved soil and leaf nutrient concentrations. Yield increases were formulation-dependent, with NPKMg 12-12-17-2 achieving a 90.5% increase, followed by NPK 15-15-15 (60.4%) and NPK 20-10-10 (30.0%) compared to the control. These findings collectively highlight the potential of targeted NPK formulations in enhancing vegetable crop performance, although the effectiveness may vary based on nutrient composition and crop type.

The application of biodigestate (BD) significantly enhanced the growth and yield of okra and cucumber compared to the control. This improvement is attributed to BD's rich nutrient profile and gradual nutrient release. During anaerobic digestion, the carbon-to-nitrogen (C:N) ratio decreases as carbon is converted to biomethane, concentrating nitrogen and minerals. Nitrogen is converted into ammonium (NH₄⁺), a plant-available form that is readily absorbed (Crolla et al., 2013). The control soil exhibited low nutrient levels, whereas BD application improved crop growth and yield by increasing soil nutrient content and availability.

BD provides essential nutrients such as N, P, K, and micronutrients (Valentinuzzi et al., 2020) and contributes organic matter that gradually releases these nutrients, ensuring a continuous supply throughout the growing season (van Midden et al., 2023). Its organic matter improves soil

structure, aeration, and water-holding capacity (Pastorelli et al., 2024), which promote root growth and enhance plant access to both nutrients and water. BD also introduces beneficial microbes (van Midden et al., 2023) that decompose organic matter, mineralize nutrients, and improve soil fertility. Similar findings were reported by Debebe and Itana (2016), who observed increased cabbage yield with bio-slurry, and Glowacka et al. (2020), who found that BD enhanced switchgrass forage yield.

Poultry manure (PM) alone outperformed both BD and NPK fertilizer. PM contains macro- and micronutrients, including N, P, K, Ca, Mg, Zn, and Cu, which are released gradually, providing sustained nutrient availability (OMEX, 2023). Compared with BD, PM contains more organic-bound phosphorus and potassium, crucial for flowering and fruiting. PM also improves soil aggregation, porosity, and water-holding capacity in sandy loam soils typical of Alfisols (Agbede, 2025). These physical improvements reduce leaching and benefit shallow-rooted crops like cucumber. PM also enriches soil with beneficial microbes that mineralize nutrients, solubilize phosphorus, and enhance nutrient uptake while promoting plant growth-promoting rhizobacteria (PGPR) that suppress soil-borne pathogens (Chen et al., 2024). Additionally, PM moderates soil pH through its basic cations, creating favorable conditions for nutrient availability and microbial activity (Arnall, 2024; Agbede, 2021). Unlike BD, which can have variable pH depending on feedstock and fermentation (Chojnacka and Moustakas, 2024), PM typically raises soil pH slightly, further enhancing nutrient use efficiency. Agbede (2021) reported that PM improved soil structure, nutrient retention, and carrot yield more effectively than NPK fertilizer. Möller and Müller (2012) noted that BD has lower organic matter content, limiting its capacity to improve soil physical properties compared with PM.

Although NPK fertilizer provides higher concentrations of N, P, and K, its effect on okra and cucumber yield is often limited due to nutrient losses—particularly nitrate leaching in light-textured soils under irrigation. This reduces nutrient use efficiency and fails to sustain growth throughout the cropping cycle (Ayoola and Adeniyi, 2006; Muhammad et al., 2022; Jia et al., 2014; Titirmare et al., 2023). In contrast, PM and BD provide a more synchronized nutrient supply through gradual mineralization, reducing leaching risk and supporting consistent crop growth (Ogboghodo et al., 2004). Their organic matter also enhances cation exchange capacity (CEC), water retention, and microbial activity, improving nutrient retention and uptake efficiency.

Integrated applications (BD+PM and BD+NPK) outperformed sole applications. These combinations improved nutrient availability and soil health, consistent with findings by Dinka et al. (2018) and Naiji Souri (2018), who reported that integrated nutrient management produced higher crop yields and improved long-term soil productivity. PM combined with NPK 15-15-15 enhances nutrient use efficiency by increasing soil organic matter and microbial activity, which boost CEC and nutrient retention (Souri et al., 2019). Studies have shown that combining organic and inorganic amendments increases yields by up to 29% compared with sole organic applications and by 8% compared with sole fertilizer applications in cereals (Weia et al., 2016).

The NPK+PM combination produced high okra and cucumber yields comparable to BD+PM. This is due to the synergy between the immediate nutrient availability from NPK and the soil-enhancing, slow-release properties of PM. PM's humus helps retain NPK-derived nutrients, reducing losses and improving uptake. Synchronizing nutrient release with crop demand is critical (Palm et al., 1997), and this combination addresses the challenge effectively. Muhammad

and Sanda (2019) found that a 50:50 PM–NPK combination significantly increased okra yield at a recommended N rate of 120 kg ha⁻¹, while Sadiq et al. (2020) reported 48.3% and 62.3% yield increases over NPK alone and control, respectively.

BD+PM outperformed BD+NPK due to the synergistic benefits of BD's readily available nutrients and PM's slow-release organic matter. BD provides soluble N, P, K, and beneficial microbes, while PM contributes humus and organic-bound nutrients that improve long-term soil fertility (Chojnacka and Moustakas, 2024; Saliga III and Skelly, 2013). This combination likely enhanced soil enzyme activity, including phosphatases and N-cycling enzymes (Lehmann et al., 2011), increasing nutrient mineralization and plant uptake (Coleman et al., 2018). In contrast, BD+NPK is more susceptible to nutrient leaching because BD lacks the humus needed to retain nutrients. Similar synergistic benefits of combined organic amendments have been reported in tomato production (Adekiya, 2019) and in studies by Li et al. (2025), which demonstrated that organic amendments improve crop growth and soil health more effectively than sole applications.

The use of BD, PM, and NPK fertilizers, whether alone or in combination, enhanced the mineral content of okra and cucumber compared to the control. This improvement resulted from increased nutrient availability and absorption. Among the sole applications, PM resulted in the highest levels of K, Ca, Fe, Zn, and Cu, which correlated with the improved soil chemical properties under this treatment. The BD+PM combination also showed high nutrient values due to the balanced soil chemical environment it provided. BD+PM encompass a wide range of macro- and micronutrients, whereas BD+NPK and NPK+PM primarily supply N, P, and K.

Consequently, the mineral composition of the soil directly affects the nutrient content of the plants.

Conclusion

This study evaluated the effects of biodigestate (BD), poultry manure (PM), and NPK fertilizer—applied individually and in combination—on soil chemical properties, growth, yield, and nutrient content of okra (*Abelmoschus esculentus*) and cucumber (*Cucumis sativus*) under controlled conditions. All soil amendments significantly improved soil fertility parameters, plant growth, and crop yield compared to the untreated control. Among individual treatments, poultry manure was the most effective in enhancing soil nutrient status and crop performance. Among the combined treatments, NPK+PM produced the highest yield and best soil fertility metrics, while BD+PM led to comparable yield and significantly higher mineral content in the fruits.

These results suggest that BD+PM can serve as a viable alternative to NPK+PM, offering comparable productivity while contributing to improved nutritional quality and long-term soil health. The use of organic-based inputs like BD and PM also aligns with global efforts toward environmentally sustainable agriculture by reducing dependence on synthetic fertilizers, minimizing nutrient leaching, and improving soil organic matter. From a policy perspective, promoting the integration of locally available organic amendments in crop production systems can reduce input costs for smallholder farmers, support circular bioeconomy practices, and contribute to national strategies on climate-smart and sustainable agriculture. Further research should investigate the long-term environmental and economic impacts of these fertilization strategies, including effects on soil carbon dynamics, post-harvest quality, and profitability.

Declaration of Competing Interest

Authors declare no competing interest whatsoever

Data availability

Data will be made available on request.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used no generative AI and AI-assisted technologies. The authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

CRedit authorship contribution statement:

AOA: Conceptualization, Original draft preparation, Manuscript editing; **SOD:** Manuscript editing; **TOO:** Supervision, analysis, Manuscript editing; **TMA:** Original draft preparation, Manuscript editing; **ODA, P-JAO:** Original draft preparation, Manuscript editing.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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